

Finite-horizon Weil coercivity and shifted-zeta contraction

Two-sided certificates for the finite-window Weil form through total horizon $\log 7$

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Abstract

We study the central Weil form $Q_{0,L}$ on complex test functions supported in an interval of total length L , together with the causal finite-horizon transfer $V_{\omega,L}$ of Suzuki’s shifted zeta ratio. A computer-assisted argument that uses no information about zeta zeros encloses the bottom of the spectrum of $Q_{0,L}$ from both sides at the seven horizons $L = \log 2, \log 3, \log 4, \log 5, \log 6, 9/5, \log 7$; in particular

$$1.37 \times 10^{-28} \leq \lambda_{\min}(Q_{0,\log 7}) \leq 6.81 \times 10^{-28},$$

so that $Q_{0,L} \succeq 1.37 \times 10^{-28} I$ for every $0 < L \leq \log 7$. The horizon $\log 7$ lies beyond $\log 6$: the generator contains the prime-power delays 2, 3, 4, 5, and the transfer contains the first mixed-prime composition $6 = 2 \cdot 3$. Lower bounds combine an analytic lower bound on the infinite Legendre tail, obtained by fractional integration, with full-output Gram matrices that retain the exact reflection-parity coupling; the resulting 64×64 Schur inequalities are verified in ball arithmetic with an explicit analytic profile remainder. Upper bounds are ball Rayleigh quotients of retained polynomial vectors. At every certified horizon the bottom of the spectrum lies in the even reflection sector, and across the sampled horizons the two-sided enclosures exhibit a rapidly increasing logarithmic decay rate, consistent with the empirical Landau–Widom picture; no decay law at unbounded depth is proved. An even-in-shift generator identity converts each floor into a small-shift contraction, for instance $\|V_{\omega,L}\| \leq \exp(-6 \times 10^{-29}\omega)$ for $0 < \omega \leq 4 \times 10^{-15}$ and $L \leq \log 7$, with quantitative reflection-positivity and cumulative-storage bounds. A second implementation reproduces every certificate, and the matrices were checked against the Fourier-side definition of the form. The results are finite-depth statements: they give neither a zero-free region nor an all-depth positivity theorem, and we estimate the rapidly increasing computational cost of extending the present scalar tail bound to larger horizons.

Keywords. Weil positivity; Riemann zeta function; causal convolution; Hankel operators; Schur complement; validated computation.

Status. Working draft. The analytic argument and the executable certificates are supplied for review; the finite calculations, their replay, an independent implementation and a Fourier-side check of the matrices have passed, but human verification of every proof and proof-assistant formalization have not been completed. The numerical results are isolated in Theorem 6.1 and Proposition 6.2.

1 Introduction and main result

Weil’s positivity criterion expresses the Riemann hypothesis as positivity of an explicit quadratic form on all compactly supported test functions. Fixing a bounded support produces a separate, finite-horizon question: is the restricted form coercive, and can an explicit positive lower bound be

proved without information about zeta zeros? Such a result controls only a bounded part of the criterion, but provides a test of methods intended to extend to larger arithmetic depth.

This paper addresses that question using the causal transfer associated with the shifted quotient

$$\Theta_\omega(z) = \frac{\xi(\frac{1}{2} - \omega - iz)}{\xi(\frac{1}{2} + \omega - iz)}, \quad \xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (1.1)$$

Suzuki introduced this family and its arithmetic Hankel kernel [6]. Our statements concern the finite causal convolution defined in Section 2; no global innerness assumption is made.

Throughout the paper, L is the *total length* of the input interval. Its centered version is $(-L/2, L/2)$, and its autocorrelation is supported in $[-L, L]$. The corresponding multiplicative cutoff is $a = e^{L/2}$. This convention is important when comparing arithmetic thresholds or compact-window bounds.

Theorem 1.1 (Two-sided finite-horizon bounds and contraction). *For each row of Table 1, with horizon L_j , floor m_j , upper bound μ_j and constants C_j, h_j, d_j , the central form (2.6) satisfies*

$$m_j \leq \lambda_{\min}(Q_{0,L_j}) \leq \mu_j, \quad \lambda_{\min}(Q_{0,L}) := \inf\{Q_{0,L}[f] : f \in \mathcal{D}_{\log,L}, \|f\|_2 = 1\}, \quad (1.2)$$

and the lower bound holds in each reflection sector separately, with the sector-wise floors and upper bounds of Table 2. Consequently, for every $0 < L \leq L_j$,

$$Q_{0,L}[f] \geq m_j \|f\|_2^2 \quad (f \in \mathcal{D}_{\log,L}). \quad (1.3)$$

For the transfer $V_{\omega,L}$ of Definition 2.1,

$$\|V_{\omega,L}\| \leq e^{-d_j\omega} < 1, \quad 0 < \omega \leq h_j, \quad 0 < L \leq L_j, \quad (1.4)$$

and $H_{\omega,L} = V_{\omega,L}R_L$ is self-adjoint with

$$I \pm H_{\omega,L} \succeq (1 - e^{-d_j\omega})I. \quad (1.5)$$

The bounds (1.2) are computer-assisted (Theorem 6.1 and Proposition 6.2); the remaining implications are analytic.

Table 1: Certified two-sided bounds and continuation constants (Theorem 1.1). m_j is an exact rational passing the full-operator test in both reflection sectors; μ_j is a certified upper bound (upper endpoint of the ball evaluation, rounded up). C_j is an integer exceeding the generator constant C_{L_j} of (3.7), and h_j, d_j are exact decimals satisfying $m_j - C_j h_j^2/3 \geq d_j$.

L_j	value	delays	floor m_j	upper μ_j	C_j	shift h_j	decay d_j
log 2	0.693	none	1.32×10^{-3}	1.33×10^{-3}	1	4×10^{-2}	6×10^{-4}
log 3	1.099	2	5.52×10^{-8}	5.54×10^{-8}	2	2×10^{-4}	2×10^{-8}
log 4	1.386	2, 3	7.48×10^{-13}	7.57×10^{-13}	4	5×10^{-7}	3×10^{-13}
log 5	1.609	2, 3, 4	8.95×10^{-18}	9.3×10^{-18}	7	1×10^{-9}	4×10^{-18}
log 6	1.792	2, 3, 4, 5	5.93×10^{-23}	7.33×10^{-23}	11	3×10^{-12}	2×10^{-23}
9/5	1.800	2, 3, 4, 5	3.38×10^{-23}	4.14×10^{-23}	11	2×10^{-12}	1×10^{-23}
log 7	1.946	2, 3, 4, 5	1.37×10^{-28}	6.81×10^{-28}	14	4×10^{-15}	6×10^{-29}

In particular $Q_{0,L} \succeq 1.37 \times 10^{-28}I$ for $L \leq \log 7$ and $Q_{0,L} \succeq 3.38 \times 10^{-23}I$ for $L \leq 9/5$. The total horizons 9/5 and log 7 are strictly larger than log 6. There the logarithmic generator has

Table 2: Sector-wise enclosures. Floors are exact rationals passing the sector test; upper bounds are ball Rayleigh quotients plus the profile error, upper endpoints rounded up. In every row the even upper bound lies below the odd floor, so the bottom of the spectrum lies in the even sector.

Horizon	even floor	even upper bound	odd floor	odd upper bound
$\log 2$	1.32×10^{-3}	1.33×10^{-3}	7.31×10^{-2}	7.32×10^{-2}
$\log 3$	5.52×10^{-8}	5.54×10^{-8}	1.49×10^{-5}	1.5×10^{-5}
$\log 4$	7.48×10^{-13}	7.57×10^{-13}	4.71×10^{-10}	4.77×10^{-10}
$\log 5$	8.95×10^{-18}	9.3×10^{-18}	9.45×10^{-15}	9.7×10^{-15}
$\log 6$	5.93×10^{-23}	7.33×10^{-23}	1.07×10^{-19}	1.36×10^{-19}
$9/5$	3.38×10^{-23}	4.14×10^{-23}	5.73×10^{-20}	7.23×10^{-20}
$\log 7$	1.37×10^{-28}	6.81×10^{-28}	4.78×10^{-25}	1.57×10^{-24}

precisely the delays $\log 2, \log 3, \log 4, \log 5$, while the coefficient of $n = 6$ occurs in the transfer as a product of two prime factors, although $\Lambda(6) = 0$; in the full-output calculation the ordered mixed overlaps $(2, 3)$ and $(3, 2)$ are retained explicitly. At the exact endpoint $L = \log 7$ the delay $T_{\log 7}$ vanishes on $(0, L)$, so $\log 7$ is the largest horizon with this generator.

Three features of Table 1 deserve comment. First, the enclosures are tight: the ratio μ_j/m_j is below 1.3 through $9/5$ and below 5 at $\log 7$, so the floors are within a small factor of the true bottom of the spectrum, and the smallness of the constants is a property of the form, not of the method. Second, the enclosures fall rapidly along the profile, at an increasing logarithmic rate: $-\log_{10} \mu_j$ grows by about 11 per unit horizon between $\log 2$ and $\log 3$ and by about 33 per unit horizon between $9/5$ and $\log 7$. These finite-difference slopes are observations on seven horizons. They prove no decay law as $L \rightarrow \infty$, which is exactly the regime that remains open; they are consistent with the empirical Landau–Widom law that Zhu [9] fits to his data and formulates as a conjecture, and whose only proved large-window form is conditional on RH. Third, the shift intervals h_j are fixed by m_j and the quadratic perturbation bound of Lemma 3.2, $h_j \approx \sqrt{3m_j/2C_j}$, and therefore shrink with the floors along the sampled horizons; only at $\log 2$, where no prime term is present, is the interval macroscopic.

The proof has three components. First, differentiating the finite Euler product gives a generator even in ω and a bounded perturbation $Q_{\omega,L} - Q_{0,L} = O_L(\omega^2)$. Second, a fractional-integration estimate gives a lower bound on the infinite Legendre tail. Third, full-output Grams control the coupling between that tail and the retained space, separately in each reflection sector. The finite comparison is a Schur inequality at the claimed positive level m_j ; positivity of a finite compression alone would not suffice. Upper bounds need only the retained head.

1.1 Relation to earlier work

Positivity of the Weil form on windows of total length at most $\log 2$, where no prime term is present, is classical, with the following hypotheses. Yoshida [8] proved, in the formulation quoted in the introduction of [1], that the Weil functional is nonnegative on smooth positive-definite functions supported in $(1/2, 2)$ whose Fourier transform vanishes at $\pm i/2$. Connes and Consani [1, Theorem 1] prove, for g supported in $[2^{-1/2}, 2^{1/2}]$ with Fourier transform vanishing at $i/2$ and at 0, the stronger inequality bounding the functional of $g * g^*$ below by the trace of a manifestly nonnegative operator built from the compressed scaling action, as part of the operator-theoretic program pursued with Moscovici and van Suijlekom. In the coordinates of Section 2 the support condition is total length $\log 2$, and the vanishing conditions at $\pm i/2$ remove the pole contribution of the explicit formula,

which in the normalization of (2.6) is the indefinite term $2|C(f)|^2 - 2|S(f)|^2$. Our log 2 row concerns the full form (2.6) on all of $\mathcal{D}_{\log, \log 2}$, pole term included and with no vanishing condition. On test functions for which the pole term vanishes, (2.6) reduces to the pole-free functional, so there the row restricts, up to the normalization of the functional, to a quantitative version of the classical statement; elsewhere it is a different statement, and we have not matched the two normalizations term by term. Beyond log 2 prime terms enter, and the positivity statements we are aware of are computer assisted. Zhu [9] certifies $Q \succeq 8.9 \times 10^{-18}$ at total length 1.6 (his window parameter is a half-width, so his $L = 0.8$ is total length 1.6 here) through a pointwise envelope for the Weil symbol and a single finite matrix, together with the certified upper bound 2.27×10^{-17} ; he observes that the window ground state is simple and even, and argues that any such one-stroke certificate must resolve frequencies up to a threshold that is doubly exponential in the horizon. Our log 5 row has the same active primes as his window and, since $\lambda_{\min}$ is nonincreasing in L , implies $\lambda_{\min}(1.6) \geq 8.95 \times 10^{-18}$ by a different reduction, consistent with his reported enclosure. The closeness of the two lower bounds is compatible evidence, not a cross-check of either implementation: lower bounds are not estimates with two-sided error bars, and the two windows differ. We do not use his certificate. The specific contributions here are the fractional tail bound of Section 4, the full-output parity reduction of Section 5, two-sided certificates at seven horizons through log 7, and the explicit positive-shift consequences. The doubly exponential threshold appears in our setting as the growth of the arithmetic loss in the scalar tail floor; Section 7.2 quantifies it. We make no claim that finite-matrix approaches to Weil positivity are themselves new.

The estimates are finite-depth. They neither establish global innerness of (1.1) nor prove a new zero-free region, and the storage corollary of Section 7 does not supply an induction in arithmetic depth.

2 Causal transfer and the central form

2.1 Finite-horizon definitions

Let $\mathcal{H}_L = L^2(0, L; \mathbb{C})$, with inner product linear in the second argument, and define

$$R_L f(x) = f(L - x), \quad T_d f(x) = \mathbf{1}_{x > d} f(x - d).$$

In particular, $T_d = 0$ for $d \geq L$ and $T_d T_e = T_{d+e}$. Every adjoint and norm below refers to unweighted $L^2(0, L)$.

Put $\Gamma_\infty(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$. For $0 < \omega \leq 1/2$ the archimedean causal convolution has Laplace transfer

$$K_\omega^\gamma(p) = \frac{\Gamma_\infty(\frac{1}{2} + p - \omega)}{\Gamma_\infty(\frac{1}{2} + p + \omega)}. \quad (2.1)$$

It may be defined on a sufficiently far right Laplace line, followed by restriction to $(0, L)$. For an explicit impulse, set $\alpha = \frac{1}{2} - \omega$, $\beta = \frac{1}{2} + \omega$. Then

$$K_\omega^\gamma(p) = \pi^\omega \frac{(p + \alpha)(p - \beta)}{(p - \alpha)(p + \beta)} \frac{\Gamma((p + \alpha)/2)}{\Gamma((p + \beta)/2)}. \quad (2.2)$$

The last ratio is the transform of $2e^{-\alpha t}(1 - e^{-2t})^{\omega-1}/\Gamma(\omega)$. The rational factor has an impulse consisting of a delta mass and smooth exponentials (with the usual limiting interpretation at $\alpha = 0$). Thus the full impulse κ_ω is locally integrable, behaves as $O(t^{\omega-1})$ at the origin, and defines a bounded finite-horizon convolution $V_{\omega, L}^\gamma$.

Definition 2.1. For $0 < \omega \leq 1/2$, define

$$b_\omega(n) = n^{\omega-1/2} \prod_{p|n} (1 - p^{-2\omega}), \quad b_\omega(1) = 1, \quad (2.3)$$

$$B_{\omega,L} = \sum_{\log n < L} b_\omega(n) T_{\log n}, \quad V_{\omega,L} = B_{\omega,L} V_{\omega,L}^\gamma. \quad (2.4)$$

Set $V_{0,L} = I$ in the strong-limit sense proved below.

The sum is finite and contains all integers below e^L . This is the logarithmic-coordinate form of Suzuki's arithmetic kernel. The Euler product identity for its Dirichlet series also identifies the full Laplace transfer, on a sufficiently far right line, as $\xi(\frac{1}{2} + p - \omega)/\xi(\frac{1}{2} + p + \omega)$. Real causal convolution gives $V_{\omega,L}^* = R_L V_{\omega,L} R_L$; therefore $H_{\omega,L} = V_{\omega,L} R_L$ is self-adjoint and $\|H_{\omega,L}\| = \|V_{\omega,L}\|$.

2.2 The central Weil form

For $f \in \mathcal{H}_L$, let $f_c(x) = f(x + L/2)$ on $(-L/2, L/2)$ and zero elsewhere. Write

$$\widehat{f}_c(\tau) = \int_{\mathbb{R}} f_c(x) e^{-i\tau x} dx, \quad C(f) = \int_{\mathbb{R}} f_c(x) \cosh(x/2) dx, \quad S(f) = \int_{\mathbb{R}} f_c(x) \sinh(x/2) dx.$$

The logarithmic form domain is

$$\mathcal{D}_{\log,L} = \left\{ f \in \mathcal{H}_L : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{f}_c(\tau)|^2 d\tau < \infty \right\}. \quad (2.5)$$

With $\psi = \Gamma'/\Gamma$, define

$$\begin{aligned} Q_{0,L}[f] &= \frac{1}{2\pi} \int_{\mathbb{R}} (\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi) |\widehat{f}_c(\tau)|^2 d\tau \\ &\quad + 2|C(f)|^2 - 2|S(f)|^2 - \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (2.6)$$

Here $\Lambda(p^k) = \log p$ and is zero otherwise. The last scalar is real. The Fourier multiplier differs from $\log(2 + |\tau|)$ by a bounded function, by the digamma asymptotics [4, §5.11]. All remaining terms are bounded. Hence (2.6) is a closed semibounded form on (2.5). It is the central geometric Weil form in this normalization. For example, if $g = f_c * \widetilde{f}_c$ with $\widetilde{f}_c(x) = \overline{f_c(-x)}$, its pole contribution is $\int g(t)(e^{t/2} + e^{-t/2}) dt = 2|C(f)|^2 - 2|S(f)|^2$. Its arithmetic contribution is $-\sum \Lambda(n)n^{-1/2}(g(\log n) + g(-\log n))$, exactly the delay term above. This also makes translation invariance explicit for complex tests. It commutes with reflection and with complex conjugation. The same notation $Q_{0,L}$ denotes its associated self-adjoint operator where appropriate.

Lemma 2.2 (Polynomial core). *Polynomials restricted to $(0, L)$ form a core for $Q_{0,L}$. If P_N is the orthogonal projection onto the polynomials of degree less than N , polynomial combinations in $(I - P_N)\mathcal{H}_L$ are a form core for the restriction to that subspace.*

Proof. Equip (2.5) with its weighted Fourier norm, equivalent to a shifted form norm. First shrink the centered zero extension by a dilation about the origin so that its support lies in a smaller interval. Dilations tending to the identity converge in this weighted norm: their norms are uniformly bounded, since the logarithmic weights at dilated frequencies are comparable, and strong convergence follows first on smooth Fourier data and then by density. Mollification with support smaller than the

resulting boundary gap converges in the weighted norm by dominated convergence. Consequently $C_c^\infty(0, L)$ is a form core.

For $h \in C^1[0, L]$, integration by parts in its zero extension gives

$$|\widehat{h}(\tau)| \leq L\|h\|_\infty, \quad |\widehat{h}(\tau)| \leq \frac{2\|h\|_\infty + L\|h'\|_\infty}{|\tau|} \quad (\tau \neq 0).$$

These bounds imply continuity from $C^1[0, L]$ to the logarithmic form domain. Approximate the derivative of a smooth function uniformly by a polynomial and integrate, matching one value. This gives polynomial approximation in C^1 , hence in form norm. Endpoint jumps of the zero extension cause no difficulty.

Finally P_N is bounded in form norm: it has finite rank, its range consists of form-domain polynomials, and its coefficient functionals are bounded in L^2 . Apply $I - P_N$ to the polynomial approximants of a vector orthogonal to the retained space. $\square$

3 The prime-delay generator and continuation

3.1 Exact logarithmic differentiation

Write $\ell(p) = \log \Gamma_\infty(\frac{1}{2} + p)$ on a far right half-plane. The archimedean logarithmic generator has symbol

$$\begin{aligned} a_\omega^\gamma(p) &= \ell'(p - \omega) + \ell'(p + \omega) \\ &= \frac{1}{p + \alpha} + \frac{1}{p - \alpha} + \frac{1}{p + \beta} + \frac{1}{p - \beta} - \log \pi \\ &\quad + \frac{1}{2}\psi((p + \alpha)/2) + \frac{1}{2}\psi((p + \beta)/2). \end{aligned} \quad (3.1)$$

It defines a causal logarithmic operator $A_{\omega, L}^\gamma$ on its natural core. Its symmetric form at zero is the gamma part of (2.6). For example, the two central rational terms give the symmetric kernel $2 \cosh((x - y)/2)$, whose form is $2|C|^2 - 2|S|^2$; the digamma term gives the multiplier in (2.6).

Proposition 3.1 (Prime-power generator). *On the causal generator core,*

$$\partial_\omega V_{\omega, L} = -A_{\omega, L} V_{\omega, L}, \quad (3.2)$$

$$A_{\omega, L} = A_{\omega, L}^\gamma - 2 \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \cosh(\omega \log n) T_{\log n}. \quad (3.3)$$

Thus its symmetric form is

$$Q_{\omega, L} = Q_{\omega, L}^\gamma - \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \cosh(\omega \log n) (T_{\log n} + T_{\log n}^*). \quad (3.4)$$

Proof. The commuting nilpotent delay algebra gives

$$B_{\omega, L} = \prod_{\log p < L} (I - p^{-1/2 - \omega} T_{\log p}) (I - p^{-1/2 + \omega} T_{\log p})^{-1}.$$

Each inverse is a terminating geometric series. Expanding the logarithm of one factor, the coefficient of $T_{k \log p}$ is $(p^{-k/2 + k\omega} - p^{-k/2 - k\omega})/k$. Its derivative is $2(\log p)p^{-k/2} \cosh(k\omega \log p)$. Differentiating the product in (2.4) and using commutation with causal gamma convolution proves (3.3). Symmetrization gives (3.4). Only the bounded arithmetic factor has been inverted; no bounded inverse of the smoothing transfer is asserted. $\square$

3.2 A bounded quadratic change in the shift

Set

$$r(t) = e^{t/2} - \frac{e^{-5t/2}}{1 - e^{-2t}}, \quad t > 0. \quad (3.5)$$

Lemma 3.2 (Bounded shift difference). *The forms $Q_{\omega,L}$, $0 \leq \omega \leq 1/2$, have the common domain $\mathcal{D}_{\log,L}$. Their gamma difference has symmetric kernel*

$$(\cosh(\omega|x-y|) - 1)r(|x-y|).$$

In particular,

$$\|Q_{\omega,L} - Q_{0,L}\| \leq C_L \omega^2, \quad (3.6)$$

as a bound on the bounded form difference, where

$$\begin{aligned} C_L = & \cosh(L/2) \left(e^{L/2} \frac{L^3}{3} + \frac{L^2}{4} \right) \\ & + \frac{1}{2} \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} (\log n)^2 \cosh(\log n/2) \|T_{\log n} + T_{\log n}^*\|. \end{aligned} \quad (3.7)$$

Proof. In the digamma integral [4, §5.9], form $a_\omega^\gamma - a_0^\gamma$ before separating the divergent terms. Its digamma part has inverse Laplace kernel $-2e^{-t/2}(\cosh(\omega t) - 1)/(1 - e^{-2t})$. The rational terms contribute $4 \cosh(t/2)(\cosh(\omega t) - 1)$. Since $2 \cosh(t/2) - e^{-t/2}/(1 - e^{-2t}) = r(t)$, symmetrization gives the asserted kernel. It is $O(|x-y|)$ at the diagonal and extends continuously in ω through $1/2$. This physical-space identity avoids taking an incomplete pointwise digamma limit at the endpoint.

Now $|r(t)| \leq e^{t/2} + 1/(2t)$ and $0 \leq \cosh(\omega t) - 1 \leq \omega^2 t^2 \cosh(t/2)/2$. The row-integral bound for the symmetric kernel is at most $\omega^2 \cosh(L/2) \int_0^L (t^2 e^{L/2} + t/2) dt$. Bounding each arithmetic difference in (3.4) in the same way gives (3.7). Bounded perturbations preserve the closed-form domain. $\square$

Lemma 3.3 (Laplace-line estimates). *Fix $\eta > 1$ and write $p = \eta + i\tau$, $\tau \in \mathbb{R}$. For $0 \leq s \leq \frac{1}{2}$ let*

$$K_s(p) = \frac{\xi(\frac{1}{2} + p - s)}{\xi(\frac{1}{2} + p + s)}, \quad a_s(p) = -\partial_s \log K_s(p).$$

Then $a_s(p) = a_s^\gamma(p) - 2 \sum_{n \geq 1} \Lambda(n) n^{-1/2} \cosh(s \log n) n^{-p}$ with an absolutely convergent series, $K_0 \equiv 1$, and there is a constant C_η such that for all $0 \leq s \leq \frac{1}{2}$ and $j \in \{0, 1, 2\}$

$$|\partial_s^j K_s(p)| \leq C_\eta (1 + \log(2 + |\tau|))^j (2 + |\tau|)^{-s}, \quad |a_s(p)| \leq C_\eta (1 + \log(2 + |\tau|)). \quad (3.8)$$

Proof. Write $K_s = K_s^\gamma K_s^\zeta$ with $K_s^\zeta(p) = \zeta(\frac{1}{2} + p - s)/\zeta(\frac{1}{2} + p + s)$. Since $\Re(\frac{1}{2} + p \mp s) \geq \eta > 1$, both zeta values are absolutely convergent Dirichlet series, bounded above and below in modulus uniformly in τ and s , and their s -derivatives are Dirichlet series with coefficients $O(\log^j n)$, also absolutely convergent and bounded. The logarithmic derivative gives the displayed arithmetic part of a_s . For the gamma factor, (2.2) gives $K_s^\gamma = \pi^s \rho_s(p) \Gamma(\frac{p+\alpha}{2})/\Gamma(\frac{p+\beta}{2})$ with the rational factor $\rho_s(p) = (p+\alpha)(p-\beta)/((p-\alpha)(p+\beta))$, where $\alpha = \frac{1}{2} - s \in [0, \frac{1}{2}]$ and $\beta = \frac{1}{2} + s \in [\frac{1}{2}, 1]$. Its denominator factors satisfy $|p-\alpha| \geq \Re(p-\alpha) \geq \eta - \frac{1}{2}$ and $|p+\beta| \geq \eta + \frac{1}{2}$ uniformly on the line (the numerator factor $p-\beta$ can be as small as $\eta - 1$ in modulus, which is harmless), so ρ_s is bounded with bounded s -derivatives. By the uniform Stirling expansion [4, §5.11(iii)], $|\Gamma(z+a)/\Gamma(z+b)|$ is comparable to $|z|^{\Re(a-b)}$ uniformly for $z = p/2$ on the line and a, b in a compact

set; here $\Re(a - b) = -s$, so the gamma ratio is comparable to $(2 + |\tau|)^{-s}$. Its s -derivatives are the ratio times polynomials in $\frac{1}{2}\psi(\frac{p+\alpha}{2})$, $\frac{1}{2}\psi(\frac{p+\beta}{2})$ and their derivatives, each $O(\log(2 + |\tau|))$ by [4, §5.11(i)]. Combining the factors gives (3.8); $K_0 = 1$ is immediate. $\square$

Proposition 3.4 (Central coercivity implies contraction). *If $Q_{0,L} \succeq mI$, then*

$$\|V_{\omega,L}\| \leq \exp(-m\omega + C_L\omega^3/3), \quad 0 < \omega \leq 1/2. \quad (3.9)$$

Proof. We give the argument in detail because the generator is unbounded and $V_{0,L} = I$ is a strong, not a norm, limit.

Weighted realization. Fix $\eta > 1$ and let M_η be multiplication by $e^{-\eta x}$, a bounded operator on $\mathcal{H}_L$ with bounded inverse. For a distribution kernel k supported in $[0, \infty)$ and g supported in $[0, L]$, $M_\eta(k * g) = (M_\eta k) * (M_\eta g)$, and multiplication by the indicator P_L of $(0, L)$ commutes with M_η . Hence the compression to $(0, L)$ of the causal operator with Laplace symbol $\kappa(p)$ equals $M_\eta^{-1} P_L \mathcal{M}[\kappa(\eta + i \cdot)] P_L M_\eta$, where $\mathcal{M}[\phi]$ denotes the Fourier multiplier by ϕ on $L^2(\mathbb{R})$. In particular $V_{s,L} = M_\eta^{-1} P_L \mathcal{M}[K_s(\eta + i \cdot)] P_L M_\eta$, and by Lemma 3.3 $\|V_{s,L}\| \leq C_\eta \|M_\eta\| \|M_\eta^{-1}\|$ uniformly in s .

Norm differentiability for positive shift. Fix $\varepsilon > 0$. By (3.8) with $j = 2$, $|K_{s+h}(p) - K_s(p) - h \partial_s K_s(p)| \leq \frac{h^2}{2} C_\eta (1 + \log(2 + |\tau|))^2 (2 + |\tau|)^{-\varepsilon} \leq C_{\eta,\varepsilon} h^2$ uniformly in τ for $s, s+h \in [\varepsilon, \frac{1}{2}]$. Therefore $s \mapsto V_{s,L}$ is norm- C^1 on $[\varepsilon, \frac{1}{2}]$ with

$$\partial_s V_{s,L} = M_\eta^{-1} P_L \mathcal{M}[\partial_s K_s(\eta + i \cdot)] P_L M_\eta = -M_\eta^{-1} P_L \mathcal{M}[(a_s K_s)(\eta + i \cdot)] P_L M_\eta. \quad (3.10)$$

Smoothing and the generator. Fix $s \in [\varepsilon, \frac{1}{2}]$, $f \in \mathcal{H}_L$, and put $g = V_{s,L} f$. By (3.8) with $j = 0$, $\mathcal{M}[K_s] M_\eta f \in H^r(\mathbb{R})$ for every $r < s$. Multiplication by the indicator of an interval is bounded on $H^r(\mathbb{R})$ for $0 \leq r < \frac{1}{2}$ [5], and on functions supported in $[0, L]$ multiplication by $e^{\eta x}$ coincides with multiplication by a smooth compactly supported function, also bounded on H^r . Hence $g \in H^r(\mathbb{R})$ for every $r < \min(s, \frac{1}{2})$, g is supported in $[0, L]$, and in particular $g \in \mathcal{D}_{\log,L}$. Since $a_s K_s$ is a bounded symbol on the line, $u := M_\eta^{-1} P_L \mathcal{M}[(a_s K_s)(\eta + i \cdot)] M_\eta f$ lies in $\mathcal{H}_L$, and by the convolution identity above u is the restriction to $(0, L)$ of $k_{a_s} * g$, where k_{a_s} is the causal distribution kernel with Laplace transform a_s : indeed $k_{a_s} * P_L(k_{K_s} * f)$ differs from $(k_{a_s} * k_{K_s}) * f$ by k_{a_s} applied to a function supported in $[L, \infty)$, whose output is supported in $[L, \infty)$. Thus (3.10) reads $\partial_s V_{s,L} f = -A_{s,L} V_{s,L} f$ with $A_{s,L} g := P_L(k_{a_s} * g)$, which is the identity (3.2), and $A_{s,L} g \in \mathcal{H}_L$.

The symmetric form of the generator. We claim $\Re\langle g, A_{s,L} g \rangle = Q_{s,L}[g]$ for the vector g above. The arithmetic part of $A_{s,L}$ is the finite sum $-2 \sum_{\log n < L} \Lambda(n) n^{-1/2} \cosh(s \log n) T_{\log n}$, since delays with $\log n \geq L$ vanish on $(0, L)$; it is bounded and its real part is the arithmetic part of (3.4). The four pole terms of (3.1) have causal kernels $e^{\mp \alpha t}$, $e^{\mp \beta t}$, bounded on $(0, L)$, and their symmetrizations are the corresponding cosh kernels (at $s = 0$, the kernel $2 \cosh((x - y)/2)$ of the pole term of (2.6)). The digamma part $\frac{1}{2}\psi(\frac{p+\alpha}{2}) + \frac{1}{2}\psi(\frac{p+\beta}{2}) - \log \pi$ has a tempered causal kernel k_ψ (a multiple of δ , a finite part of $1/t$, and a bounded function); for $g \in L^2$ with compact support, $k_\psi * g$ is tempered with Fourier transform $\hat{k}_\psi(\tau) \hat{g}(\tau)$, where $\hat{k}_\psi(\tau) = \frac{1}{2}\psi(\frac{i\tau+\alpha}{2}) + \frac{1}{2}\psi(\frac{i\tau+\beta}{2}) - \log \pi$ is the Laplace symbol on the imaginary axis; for $s < \frac{1}{2}$ its arguments stay away from the poles of ψ and it is $O(\log(2 + |\tau|))$. Since $g \in H^r$ with $r > 0$, $k_\psi * g \in L^2$, and Plancherel gives $\Re\langle g, k_\psi * g \rangle = \frac{1}{2\pi} \int \Re \hat{k}_\psi(\tau) |\hat{g}(\tau)|^2 d\tau$, the digamma term of $Q_{s,L}$. Summing the three parts proves the claim; this is the extension of Proposition 3.1 from the smooth core to g , and the same computation shows that the form $Q_{s,L}$ so obtained is the one whose difference from $Q_{0,L}$ is bounded by Lemma 3.2.

Energy inequality and Gronwall. With $g_s = V_{s,L} f$, norm differentiability gives, for $s \in [\varepsilon, \frac{1}{2}]$,

$$\frac{d}{ds} \|g_s\|^2 = 2 \operatorname{Re} \langle g_s, \partial_s g_s \rangle = -2 \operatorname{Re} \langle g_s, A_{s,L} g_s \rangle = -2 Q_{s,L}[g_s] \leq -2(m - C_L s^2) \|g_s\|^2,$$

using the hypothesis and Lemma 3.2. Integrating from ε to $\omega < \frac{1}{2}$,

$$\|V_{\omega,L}f\|^2 \leq \|V_{\varepsilon,L}f\|^2 \exp\left(-2 \int_{\varepsilon}^{\omega} (m - C_L s^2) ds\right).$$

The zero-shift limit. Finally $\|V_{\varepsilon,L}f - f\| \leq \|M_{\eta}^{-1}\| \|(\mathcal{M}[K_{\varepsilon}(\eta + i\cdot)] - I)M_{\eta}f\|_{L^2(\mathbb{R})}$, and the right side tends to 0 as $\varepsilon \downarrow 0$ by Plancherel and dominated convergence, since $K_{\varepsilon}(\eta + i\tau) \rightarrow 1$ pointwise with $|K_{\varepsilon}| \leq C_{\eta}$. Letting $\varepsilon \downarrow 0$ proves (3.9) for $0 < \omega < \frac{1}{2}$; the case $\omega = \frac{1}{2}$ follows by the norm continuity of $s \mapsto V_{s,L}$ established above. No norm derivative at zero and no bounded inverse of the smoothing transfer are used. $\square$

4 An analytic lower bound on the infinite Legendre tail

Let $\phi_n(x) = \sqrt{(2n+1)/L} P_n(2x/L - 1)$ and let P_N retain $\phi_0, \dots, \phi_{N-1}$. Define causal fractional integration by

$$I_{\nu}f(x) = \frac{1}{\Gamma(\nu)} \int_0^x (x-y)^{\nu-1} f(y) dy.$$

The following estimate applies to all combinations in the infinite tail.

Lemma 4.1 (Fractional tail estimate). *For $N \geq 1$ and $0 < \nu < 1$,*

$$\|I_{\nu}(I - P_N)\| \leq (L/2)^{\nu} \sqrt{\frac{\Gamma(N+1-\nu)}{\Gamma(N+1+\nu)}}. \quad (4.1)$$

Consequently

$$\|I_1(I - P_N)\| \leq \frac{L}{2\sqrt{N(N+1)}}. \quad (4.2)$$

Proof. Termwise beta integration of the terminating Legendre polynomial gives

$$I_{\nu}P_n(2x/L - 1) = \frac{n!}{\Gamma(n+1+\nu)} x^{\nu} P_n^{(-\nu,\nu)}(2x/L - 1).$$

The polynomial and orthogonality identities may also be read from [4, §§18.3, 18.5, 18.17]. For a finite tail combination, use $x^{2\nu} \leq (L/2)^{2\nu} x^{\nu} (L-x)^{-\nu}$ and Jacobi orthogonality. Indeed $x(L-x) \leq L^2/4$, and the required Jacobi norm is

$$\int_0^L x^{\nu} (L-x)^{-\nu} [P_n^{(-\nu,\nu)}(2x/L - 1)]^2 dx = \frac{L \Gamma(n+1-\nu) \Gamma(n+1+\nu)}{(2n+1)(n!)^2}.$$

The squared norm becomes bounded by

$$(L/2)^{2\nu} \sum_{n \geq N} |c_n|^2 \frac{\Gamma(n+1-\nu)}{\Gamma(n+1+\nu)}.$$

Successive gamma ratios decrease by the factor $(n+1-\nu)/(n+1+\nu) < 1$. Density and boundedness of I_{ν} prove (4.1). Letting $\nu \uparrow 1$, using convergence of the fractional kernels in $L^1(0, L)$, gives (4.2). $\square$

Remark 4.2 (Sharpness on single modes). The bound is essentially optimal in its leading term. By the Weber–Schafheitlin integral [4, §10.22(ii)], differentiated in the exponent, the exact value of the logarithmic part on a single normalized Legendre mode is

$$\frac{1}{2\pi} \int_{\mathbb{R}} \log \frac{|\tau|}{2\pi} |\widehat{(\phi_n)_c}(\tau)|^2 d\tau = \log 2 + \frac{1}{2}(\psi(n + \frac{1}{2}) + \psi(n + \frac{3}{2})) - \log(\pi L),$$

which exceeds the floor $H_N - \gamma - \log(\pi L)$ at $n = N$ by $\log 2 + O(1/N)$. The floor must hold for all tail combinations, and the logarithmic multiplier is not diagonal in the Legendre basis, so a loss of this order is expected. The identity was also used to check the computed head matrices independently of the formulas of Section 6.

The central gamma derivative $U^\gamma = \partial_\omega V_{\omega,L}^\gamma|_0$ has the decomposition

$$U^\gamma = J_L + S_w, \quad J_L = \partial_\nu[(2\pi)^\nu I_\nu]|_0, \quad w(t) = \frac{G_0(t) - 1}{t}, \quad (4.3)$$

where S_w is causal convolution and

$$G_0(t) = \frac{e^{t/2}t}{\sinh t} - 4t \cosh(t/2) = \sum_{j \geq 0} g_j t^j. \quad (4.4)$$

To verify (4.3), the inverse kernel of $-a_0^\gamma$, away from the origin, is $e^{t/2}/\sinh t - 4 \cosh(t/2)$; subtracting the finite-part kernel of J_L leaves w . The constant term agrees with the Laplace asymptotic $-a_0^\gamma(p) = \log(2\pi/p) + O(p^{-1})$. In particular $g_0 = 1$, $g_1 = -7/2$, and $g_2 = -1/24$.

Proposition 4.3 (Central infinite-tail floor). *Suppose $K_L \geq |w(0)| + \int_0^L |w'(t)| dt$. Then, on $(I - P_N)\mathcal{D}_{\log,L}$,*

$$Q_{0,L} \succeq a_{N,L} I, \quad (4.5)$$

where

$$\begin{aligned} a_{N,L} &= H_N - \gamma - \log(\pi L) - \frac{K_L L}{2\sqrt{N(N+1)}} \\ &\quad - \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \|T_{\log n} + T_{\log n}^*\|. \end{aligned} \quad (4.6)$$

Proof. Apply (4.1) to a polynomial tail vector, multiply by $(2\pi)^\nu$, square, and differentiate at $\nu = 0$. Both sides have the same value there. The result is $-\operatorname{Re}\langle f, J_L f \rangle \geq (H_N - \gamma - \log(\pi L))\|f\|^2$. The differentiation is legitimate on each polynomial, whose derivative contains only polynomials and logarithmic endpoint terms in L^2 .

Next $S_w = (w(0)I + S_{w'})I_1$. Young's inequality and (4.2) give

$$\|S_w(I - P_N)\| \leq \frac{K_L L}{2\sqrt{N(N+1)}}.$$

Since $Q_0^\gamma = -\operatorname{Re} U^\gamma$ in form sense, this is a lower-bound loss. Bound the remaining symmetric delays individually in norm. The resulting inequality $Q_{0,L}[t] \geq a_{N,L}\|t\|^2$ holds for every polynomial tail combination t . Both sides are continuous in the form norm of $\mathcal{D}_{\log,L}$ (the left side because $Q_{0,L}$ is closed, the right side trivially), and by Lemma 2.2 polynomial combinations in $(I - P_N)\mathcal{H}_L$ are a form core for the restriction of $Q_{0,L}$ to $(I - P_N)\mathcal{D}_{\log,L}$; hence the inequality extends to the complete tail form domain. $\square$

For clarity, the delay norm in (4.6) is not automatically 2. Decompose x into $x = r + kd$, $0 < r < d$. On each fiber, $T_d + T_d^*$ is the adjacency matrix of a finite path. If $q = \lceil L/d \rceil$, its norm is

$$\|T_d + T_d^*\| = 2 \cos \frac{\pi}{q+1} \quad (d < L), \quad (4.7)$$

with threshold equality using the shorter path, since the exceptional fiber has measure zero. In particular the norm is 1 for $d < L \leq 2d$ and $\sqrt{2}$ for $2d < L \leq 3d$. The reviewed implementation evaluates the cosine formula for every finite path length. Logarithmic endpoints use exact integer-power comparisons; unresolved ball comparisons at other horizons stop the calculation.

4.1 A uniform analytic profile remainder

Lemma 4.4 (Profile enclosure). *For $0 < L < 3$ and an integer $M \geq 1$, let $\tilde{U}$ retain the terms g_j , $j \leq M$, in (4.3), leaving J_L and the arithmetic delays exact. Then the bounded difference satisfies*

$$\|U - \tilde{U}\| \leq \eta_M, \quad \eta_M = \frac{256(L/3)^{M+1}}{(M+1)(1-L/3)}. \quad (4.8)$$

One may use

$$K_L = \sum_{j=1}^M |g_j| L^{j-1} + \frac{256}{3} \frac{(L/3)^M}{1-L/3} \quad (4.9)$$

in Proposition 4.3.

Proof. On $|z| = 3$, the Euler product for $\sinh z/z$ [4, §4.36] gives $|z/\sinh z| \leq 3/\sin 3 < 24$. Since $e^{3/2} < 5$, (4.4) implies $|G_0(z)| < 5(24+12) < 256$. Thus $|g_j| \leq 256 3^{-j}$. The L^1 norm of the omitted convolution kernel is bounded by $\sum_{j>M} |g_j| L^j/j \leq \eta_M$, proving (4.8). For w , the constant term contributes $|g_1|$, while integration of each derivative term gives $|g_j| L^{j-1}$. Summing the omitted geometric series proves (4.9). $\square$

The majorant 256 is deliberately loose: numerically $\sup_{|z|=3} |G_0(z)| \approx 28.3$ and $\max_j |g_j| 3^j \approx 15.2$. Since η_M is already far below every quantity it is compared with (Table 3), nothing is gained by tightening it.

5 Exact reflection-parity coupling and the Schur test

Write $P = P_N$, let P_r be its parity- r part ($r = \pm 1$), and denote the central model form by $\tilde{Q} = -(\tilde{U} + \tilde{U}^*)/2$. The polynomial space lies in the operator domain: the fractional derivative of a polynomial has only square-integrable logarithmic endpoint terms, and all other terms are bounded convolutions or delayed polynomials. Thus $Q_0 P$ and $\tilde{Q} P$ are bounded finite-column operators.

Define the retained matrix and two full-output Grams by

$$q = P \tilde{Q} P, \quad F = (\tilde{U} P)^* (\tilde{U} P), \quad C = (\tilde{U} P)^* R_L (\tilde{U} P). \quad (5.1)$$

In this notation, F and C are finite matrices but the output in their definition has *not* been projected onto $P\mathcal{H}_L$.

Lemma 5.1 (Exact central-parity leakage). *If q_r, F_r, C_r denote the same-parity blocks in (5.1), then*

$$E_r = \frac{1}{2}(F_r + r C_r) - q_r^2 = ((I - P) \tilde{Q} P_r)^* ((I - P) \tilde{Q} P_r). \quad (5.2)$$

Proof. Causal convolution gives $\tilde{U}^* = R_L \tilde{U} R_L$. For $R_L f = r f$, $\tilde{Q} f = -(\tilde{U} f + r R_L \tilde{U} f)/2$. Taking the full-output squared norm yields $\|\tilde{Q} f\|^2 = \langle f, (F_r + r C_r) f \rangle / 2$. Subtract the retained-output norm $\|q_r f\|^2$ and polarize. $\square$

Proposition 5.2 (Certified finite test). *Suppose the exact central tail is at least aI , $a > m$, and $\|\tilde{U} P\| < K$. If both finite inequalities*

$$(a - m)(q_r - mI) - E_r - \epsilon I \succeq 0, \quad r = \pm 1, \quad (5.3)$$

hold, where

$$\epsilon = (|a - m| + 4K)\eta_M + 2\eta_M^2, \quad (5.4)$$

then $Q_{0,L} \succeq mI$ on the full form domain.

Proof. The profile difference has norm at most η_M , as does its symmetric part. On each input parity, $\tilde{Q} P_r$ is an output-parity projection of $-\tilde{U} P_r$, and hence has norm below K . The full squared-output Gram and the squared retained matrix each change by at most $2K\eta_M + \eta_M^2$. The linear head term changes by at most $|a - m|\eta_M$. Therefore (5.3) implies

$$(a - m)(Q_{P,r} - mI) - B_r^* B_r \succeq 0, \quad B_r = (I - P)Q_0 P_r,$$

for the exact central form.

For an arbitrary input in one sector, write it as $h + t$ with h retained and t in the form-domain tail. The quadratic form of $Q_0 - mI$ is bounded below by

$$\langle h, (Q_{P,r} - mI)h \rangle + 2 \operatorname{Re} \langle B_r h, t \rangle + (a - m)\|t\|^2.$$

Completing the square, or minimizing over t , leaves $\langle h, (Q_{P,r} - mI - (a - m)^{-1} B_r^* B_r)h \rangle \geq 0$. The two orthogonal reflection sectors exhaust the complex Hilbert space. $\square$

The use of the exact parity leakage is consequential. Replacing it by the full leakage of $\tilde{U}$ also bounds the coupling, but charges irrelevant output parity against the available tail margin. Equation (5.2) removes that loss without discarding any coupling of the central form.

Remark 5.3 (A smaller error for future calculations). The certificates below keep the conservative error (5.4). One can instead set

$$\epsilon_r = (|a - m| + 2\kappa_r)\eta_M + \eta_M^2, \quad \kappa_r \geq \|(I - P)\tilde{Q} P_r\|.$$

Indeed the omitted-output column operator changes in norm by at most η_M , so its squared Gram changes by at most $2\kappa_r\eta_M + \eta_M^2$ directly. There is no need to bound the full Gram and the squared head separately. A certified upper bound on $\sqrt{\operatorname{tr} E_r}$ supplies κ_r , since E_r is an exact positive semidefinite Gram. Any numerical upper bound used here must include the ball uncertainty; this refinement is not used to justify the reported runs.

6 Finite construction and the certificates

6.1 Polynomial and logarithmic output formulas

Use the unitary coordinate $u = x/L$, so the normalized basis is $\sqrt{2n+1} p_n(u)$, where

$$p_n(u) = \sum_{k=0}^n (-1)^{n+k} \binom{n}{k} \binom{n+k}{k} u^k. \quad (6.1)$$

With $\ell_L = \gamma + \log(2\pi L)$, the gamma derivative of a monomial is

$$U^\gamma u^k = u^k(\log u + \ell_L - H_k) + \sum_{j \geq 1} g_j L^j \frac{(j-1)!k!}{(j+k)!} u^{j+k}. \quad (6.2)$$

This follows by differentiating the beta integral for $I_\nu u^k$ and integrating each term of S_w . For the model, the sum stops at $j = M$. Every output therefore has the form

$$\tilde{U}p_n(u) = p_n(u) \log u + q_n(u) + \sum_{\log j < L} \frac{2\Lambda(j)}{\sqrt{j}} \mathbf{1}_{u > \delta_j} p_n(u - \delta_j), \quad \delta_j = \frac{\log j}{L}. \quad (6.3)$$

The polynomial q_n is supplied by (6.2). The head, F , and C are integrated using exact formulas for moments of $1, \log u, (\log u)^2, \log(1-u)$, and $\log u \log(1-u)$, including truncated integration intervals. Appendix A lists them. There is no quadrature error and no finite-output truncation.

For the reflected delayed products, the nonempty integration window is $(\delta_i, 1 - \delta_j)$, equivalently $\log(ij) < L$. At $L = 9/5$ and at $L = \log 7$ the active generator indices are 2, 3, 4, 5, and the ordered overlap pairs are exactly (2, 2), (2, 3), (3, 2); at $L = \log 5$ and $L = \log 6$ only (2, 2) is present, and there is none at $\log 4$ and below.

6.2 Enclosed arithmetic

The standalone implementation uses Arb real balls [3] through python-flint [7]. Integer and rational coefficients in (4.4), (6.1), and (6.2) are formed exactly. All subsequent conversions, elementary constants, logarithms, square roots, moments, polynomial reflections, and matrix operations are performed with balls, at 1536-bit working precision for the horizon 9/5 and 1792 bits at the logarithmic horizons. Exact symmetries are imposed only after checking consistency of the enclosures.

The matrix F encloses the model-output Gram, and its trace is checked below 10^7 . Hence $\|\tilde{U}P\| < 10000$, allowing $K = 10000$ in (5.4); the computed traces are below 2400 at every horizon, and the deliberately loose bound avoids making this part of the proof delicate. An outward lower value of (4.6) and an outward upper value of (4.8) are used. An outward upper value of (5.4) is then subtracted from each model Schur matrix.

For a symmetric ball matrix M , the sign test uses the recurrence

$$\begin{aligned} d_i &= M_{ii} - \sum_{k < i} \ell_{ik}^2 d_k, \\ \ell_{ji} &= \frac{M_{ji} - \sum_{k < i} \ell_{jk} \ell_{ik} d_k}{d_i} \quad (j > i). \end{aligned}$$

Every pivot must be strictly positive as a ball. All divisions then have denominators excluding zero. Induction encloses the exact LDL^* factors of every symmetric matrix contained in M ; positive pivots imply its positivity. Ordinary floating-point eigenvalues never enter this decision.

6.3 Certified lower bounds

Theorem 6.1 (Computer-assisted matrix inequalities). *At each horizon L_j of Table 1, with $N = 128$ and the profile degree and working precision of Table 3, both inequalities (5.3) hold at $m = m_j$ with $K = 10000$, the tail floor (4.6), and the error (5.4). In the even and odd sectors separately, (5.3) holds at the sector floors of Table 2. In particular $Q_{0,L_j} \succeq m_j I$.*

Computer-assisted proof. The supplied builder evaluates (6.3), the moment formulas in Appendix A, and the bounds of Section 4 with ball arithmetic. At the two leading horizons the following rationally rounded inequalities are checked:

$$\begin{aligned} a_{128,9/5} &> 0.69124708426711, & K_{9/5} &< 5.482684, & \eta_{180} &< 2.477 \times 10^{-40}, \\ a_{128,\log 7} &> 0.60748624181467, & K_{\log 7} &< 5.839924, & \eta_{220} &< 9.342 \times 10^{-42}; \end{aligned}$$

the arithmetic norm term in (4.6) is approximately 2.39376738699 at both, and the tail floors at all seven horizons are listed in Table 3. The sector floors were located as follows: for each sector the largest m passing the sector inequality was found by bisection with the same ball test, rounded down to three significant figures, and the resulting exact rational was tested again; m_j is the smaller of the two sector floors and was tested in both sectors. All 64×64 ball LDL tests pass; the minimum pivots are displayed in Table 3. Applying Proposition 5.2 gives the full-operator conclusion. $\square$

Table 3: Certificate data at $N = 128$. The tail floor $a_{128,L}$ is an outward lower value of (4.6) (lower endpoint, rounded down); ϵ is the outward upper value of (5.4) subtracted before each sign test (upper endpoint, rounded up); pivots are lower values of the smallest LDL pivots at the floor m_j (rounded down; summaries, not spectral bounds); the radius is the largest ball radius among the four saved matrices (upper endpoint, rounded up). “Required” is the floating-point diagnostic $\lambda_{\max}(q_r^{-1/2} E_r q_r^{-1/2})$, the tail floor the model test would need at level zero, rounded to nearest. The certified endpoints themselves are in the JSON records.

L	M	bits	$a_{128,L}$	ϵ	pivot (even)	pivot (odd)	radius	required
log 2	220	1792	4.0674	1.45×10^{-136}	2.75×10^{-2}	1.08	1.7×10^{-344}	0.166 / 0.166
log 3	220	1792	3.1089	2.8×10^{-92}	1.18×10^{-2}	1×10^{-1}	2.6×10^{-305}	0.224 / 0.225
log 4	220	1792	2.2350	6.95×10^{-70}	5.71×10^{-3}	1.32×10^{-2}	9.4×10^{-297}	0.293 / 0.289
log 5	220	1792	1.5295	1.71×10^{-55}	7.53×10^{-4}	3.64×10^{-3}	5.7×10^{-294}	0.348 / 0.340
log 6	220	1792	0.6961	3.91×10^{-45}	6.88×10^{-5}	2.01×10^{-4}	1.3×10^{-291}	0.469 / 0.473
9/5	180	1536	0.6912	9.91×10^{-36}	5.67×10^{-5}	1.81×10^{-4}	9.4×10^{-215}	0.491 / 0.480
log 7	220	1792	0.6074	3.74×10^{-37}	1.69×10^{-5}	9.03×10^{-5}	4.7×10^{-295}	0.595 / 0.590

Table 4: Lowest five eigenvalues of the 64×64 head q_r in each sector (floating-point diagnostics from the ball midpoints, rounded to nearest; not certified). The certified statements are the enclosures of Tables 1 and 2.

Horizon	sector	lowest five head eigenvalues
9/5	even	$4.14 \times 10^{-23}, 7.25 \times 10^{-17}, 1.48 \times 10^{-11}, 4.02 \times 10^{-7}, 2.77 \times 10^{-3}$
9/5	odd	$7.22 \times 10^{-20}, 4.19 \times 10^{-14}, 3.26 \times 10^{-9}, 3.7 \times 10^{-5}, 6.88 \times 10^{-2}$
log 7	even	$6.8 \times 10^{-28}, 1.66 \times 10^{-21}, 8.19 \times 10^{-16}, 6.02 \times 10^{-11}, 1.25 \times 10^{-6}$
log 7	odd	$1.57 \times 10^{-24}, 1.32 \times 10^{-18}, 3.21 \times 10^{-13}, 8.15 \times 10^{-9}, 1.09 \times 10^{-4}$

The largest radius among the saved matrix enclosures is below 10^{-214} at every horizon (Table 3). This finite-arithmetic uncertainty is propagated through the sign tests; it is separate from the analytic remainder. The matrix archive stores dyadic midpoint/radius pairs and checks that every reconstructed ball contains the original; reconstructing a radius may enlarge it outward. A separate process loaded each archive and repeated the LDL tests successfully.

6.4 Certified upper bounds

Proposition 6.2 (Upper bounds). *For each horizon and each reflection sector, $\lambda_{\min}$ of the sector form is at most the upper bound listed in Table 2; the even values are the μ_j of Table 1. At every horizon the even upper bound is smaller than the odd floor, so the bottom of the spectrum of Q_{0,L_j} lies in the even sector.*

Computer-assisted proof. The retained polynomials lie in the operator domain and $\|Q_{0,L} - \tilde{Q}\| \leq \eta_M$ (Lemma 4.4), so for every vector v in the retained space of parity r ,

$$\lambda_{\min}(Q_{0,L}|_{\text{parity } r}) \leq \frac{\langle v, q_r v \rangle}{\langle v, v \rangle} + \eta_M.$$

For each sector, a floating-point eigenvector of the ball midpoints of q_r for its smallest eigenvalue is rounded to a rational vector v with forty digits per coordinate, and the right side is evaluated with balls, using the saved enclosure of q_r and the outward value of η_M ; its upper end, rounded up, is the tabulated bound. The last assertion is the comparison of two exact numbers in each row of Table 2. $\square$

The vectors v are diagnostics; the certified content is the ball evaluation. Table 4 lists the lowest head eigenvalues in each sector at the two leading horizons. Two regularities are visible at every horizon: the spectrum alternates between the sectors, the odd ground state lying between the first two even eigenvalues, and consecutive same-parity eigenvalues are separated by factors of about $10^{5.5}$ to 10^6 at the deeper horizons. These are floating-point observations; the certified statements are the enclosures.

Proof of Theorem 1.1. Theorem 6.1 and Proposition 6.2 give (1.2) at $L = L_j$. Zero extension into a larger interval preserves the central form: the Fourier representation and its translation-invariant kernels, including the delay correlations, do not depend on the chosen containing horizon. Thus the lower bound holds at every shorter horizon, which is (1.3).

Direct ball evaluation of (3.7) gives $C_{L_j} < C_j$ for the integers C_j of Table 1; at the two leading horizons $C_{9/5} < 10.298615 < 11$ and $C_{\log 7} < 13.541169 < 14$. The exact rational inequalities $m_j - C_j h_j^2/3 \geq d_j$ are verified for every row by the supplied analyzer; for instance

$$\begin{aligned} 3.38 \times 10^{-23} - \frac{11}{3}(2 \times 10^{-12})^2 &> 1.91 \times 10^{-23} \geq 10^{-23}, \\ 1.37 \times 10^{-28} - \frac{14}{3}(4 \times 10^{-15})^2 &> 6.2 \times 10^{-29} \geq 6 \times 10^{-29}. \end{aligned}$$

Proposition 3.4 then proves (1.4) at the outer horizon L_j . Causality makes every shorter transfer a compression, so it has no larger norm. Self-adjointness of $H = VR_L$ and $\|H\| = \|V\|$ give (1.5). $\square$

6.5 What the larger head resolved

The preceding 96-mode experiment at horizon 9/5 did not pass the sufficient Schur test. To explain the improvement, Table 5 compares the scalar tail floor with the diagnostic largest eigenvalue of $q_r^{-1/2} E_r q_r^{-1/2}$. The latter measures the tail floor required by the model test at level zero. It was computed using a high-precision congruence before conversion to floating point and is a diagnostic only.

Thus increasing the retained space improved both the tail floor and the remaining coupling. Failure of the earlier sufficient inequality was not evidence of a negative direction of the full form. At $N = 128$ the even head ground state is 4.14×10^{-23} , so the certified floor 3.38×10^{-23} is within

Table 5: Head/tail comparison at $L = 9/5$. Only the final column represents the complete certified sign test.

N	Tail floor	Required, even	Required, odd	Complete test
96	0.39212903	0.55845	0.54762	Not certified
128	0.69124708	0.49054476	0.47957150	Pass at $m = 3.38 \times 10^{-23}$

a factor 1.23 of the best that any test on this head could give. The old matrix construction used a separate aggregate arithmetic budget restricted to at most 96 modes; that budget was not extended here, and the new direct-ball calculation is independent of it. As a normalization diagnostic, all shared entries of the new head and full-output Grams agree with the earlier 96-mode point matrices to within 3×10^{-217} after conversion of the causal/Hankel signs. Those old entries are not inputs to the new positivity proof.

At log 7 the comparison is tighter: the even and odd relative-coupling diagnostics are 0.5948 and 0.5899 against the available floor $a_{128, \log 7} \approx 0.6075$, ratios of 0.979 and 0.971. The 128-mode calculation therefore succeeds at this endpoint, but Table 3 shows the required floor rising and the available floor falling steadily along the profile; the two curves cross soon after log 7 (Section 7.2).

6.6 Independent reconstruction and Fourier-side checks

A second implementation, `independent_arb.py`, was begun from the printed equations before the supplied numerical source became available. It normalizes the polynomial rows before assembly and computes reflected gamma-delay integrals over $(0, 1 - \delta)$, rather than using the supplied builder’s tail moments over $(\delta, 1)$. It reproduces the full-operator tests and their displayed pivots at all seven horizons; at 9/5 and log 7 its matrices are archived, all entries of the retained head and both full-output Grams have overlapping enclosures between implementations, and replaying the independently saved Schur matrices also passes.

A third check, performed during the review of version 0.2 and included in the repository, evaluates the matrices from the Fourier-side definition (2.6) without using the formulas of Section 6: the central form of a smooth even test at $L = 9/5$ agrees with $\sum_{\rho} |\hat{f}_c(\gamma_{\rho})|^2$ over the first hundred zeta zeros to a relative 6×10^{-10} , fixing the normalization; the diagonal head entries $Q_{0,L}[\phi_n]$ for $n \leq 20$, computed with the exact logarithmic part of Remark 4.2 and quadrature for the rest, agree with the Arb head to 10^{-11} ; and full-output Gram entries computed by direct quadrature of the outputs agree with the archive to more than twenty digits. These checks share no code with the two implementations. All three routes use the same analytic reduction; none constitutes independent human review or formal verification.

7 Storage, scope, and the next depth problem

Corollary 7.1 (Cumulative storage). *Fix a row of Table 1. For any spatial split of $(0, L)$ with $L \leq L_j$, write*

$$V_{\omega, L} = \begin{pmatrix} X & 0 \\ Y & Z \end{pmatrix}.$$

For $0 < \omega \leq h_j$,

$$I - X^*X - Y^*(I - ZZ^*)^{-1}Y \succeq (1 - e^{-2d_j\omega})I. \quad (7.1)$$

Proof. Let $\rho = e^{-d_j\omega}$. Theorem 1.1 gives $I - V^*V \succeq (1 - \rho^2)I$, and $\|Z\| \leq \rho < 1$. The Schur complement of the lower-right block $I - Z^*Z$ is

$$I - X^*X - Y^*Y - Y^*Z(I - Z^*Z)^{-1}Z^*Y = I - X^*X - Y^*(I - ZZ^*)^{-1}Y.$$

For a fixed first-block vector, minimizing the defect quadratic form over the second-block vector leaves a value at least $(1 - \rho^2)$ times the first-block norm squared. This proves (7.1). $\square$

In particular, the split $\log 7 = \log 5 + \log(7/5)$ includes the first mixed-prime region. The corollary is obtained by certifying the whole larger operator first. It does not show that positivity of a previous slab alone forces the next storage update to be positive; that distinction separates the present finite-depth results from an induction in depth. The algebraic identities that such an induction would have to carry are recorded in Appendix B; they specify bookkeeping, not a mechanism.

7.1 A diagonal formulation of the all-depth target

Proposition 7.2. *Suppose there exist $L_j \rightarrow \infty$ and $\omega_j \downarrow 0$ such that $\|V_{\omega_j, L_j}\| \leq 1$ for every j . Then the central Weil form is nonnegative on every compactly supported smooth test function. By Weil's criterion, such a sequence would imply RH.*

Proof. Fix a finite horizon L and a smooth test f compactly supported inside it. For all sufficiently large j , causal compression gives $\|V_{\omega_j, L}f\| \leq \|f\|$. For such f the map $s \mapsto \|V_{s, L}f\|^2$ is differentiable at $s = 0$ with derivative $-2Q_{0, L}[f]$: in the weighted realization of the proof of Proposition 3.4, $\widehat{M_\eta}f$ is rapidly decreasing while $\partial_s K_s(\eta + i\tau)$ grows only logarithmically, so dominated convergence applies to the difference quotient on the Laplace line, and the identification of the resulting pairing with $Q_{0, L}[f]$ is the case $s = 0$ of the form identity proved there. Hence

$$\lim_{s \downarrow 0} \frac{\|f\|^2 - \|V_{s, L}f\|^2}{2s} = Q_{0, L}[f].$$

The quotients along $s = \omega_j$ are nonnegative, so their limit is nonnegative. The horizon and compact test were arbitrary; the final implication is the classical Weil criterion [1, 2]. $\square$

The proposition requires neither a depth-independent coercivity margin nor a fixed positive shift interval. Nevertheless, constructing the entire sequence remains the missing all-depth statement; the seven horizons certified here are not such a sequence, and Table 1 shows the shift intervals h_j that the present method certifies shrinking rapidly along the sampled horizons. The finite data establish no rate for that collapse at unbounded depth.

7.2 Limitations: the scale of the next horizons

The scalar tail floor (4.6) is

$$a_{N, L} \approx \log N - \log(\pi L) - \Sigma_L, \quad \Sigma_L = \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \|T_{\log n} + T_{\log n}^*\|,$$

up to the small smooth-profile term. It pays separately for every arithmetic delay norm, and Σ_L grows like a multiple of $e^{L/2}$ by the prime number theorem. What follows are resource projections for the present implementation, not impossibility bounds. Indeed the scalar floor alone never obstructs the test at a fixed horizon: for $0 < L < 3$ and any admissible profile degree M the constants K_L

and Σ_L in (4.6) are finite, so $a_{N,L} \rightarrow +\infty$ as $N \rightarrow \infty$. The retained head, the coupling, the profile error and the cost of the ball arithmetic decide whether a certificate is obtained, and the arithmetic loss governs how fast that cost grows. The floor the finite test needed at the certified horizons is the relative-coupling diagnostic of Table 3, which rises from 0.17 at $\log 2$ to 0.59 at $\log 7$ while $a_{128,L}$ falls from 4.07 to 0.61; whether a floor near 0.6 suffices at an uncomputed horizon is not known, since the required coupling has risen with every new prime. At the next prime-active horizon $\log 8$, where the generator acquires the delay $\log 7$ and the endpoint delay $\log 8$ vanishes, $\Sigma_{\log 8} \approx 3.13$ and the tail floor at $N = 128$ is negative (≈ -0.20); it is 0.52 at $N = 256$ and 1.22 at $N = 512$. Keeping the floor near 0.6 would need roughly $N \approx 310$ at $\log 8$, 950 at $\log 11$, 1900 at $\log 13$, 4400 at $\log 16$, and 27,000 as $L \rightarrow 3$, the boundary of the domain of the proved profile remainder (Lemma 4.4). A local extrapolation of Table 1 suggests that the certified margin shrinks by a factor of roughly 10^{-34} per additional unit of horizon, and the model error must be kept below it. Matrix assembly scales like $N(N + M)^2$ ball products. Extrapolating these figures, the cost of the present implementation rises rapidly beyond the certified range, and horizons near $\log 13$ to $\log 16$ would be demanding for it; none of this predicts whether a certification there succeeds or fails. The growth of Σ_L is the tail-side appearance of the doubly exponential resolution threshold that Zhu [9] derives for one-stroke certificates.

The small-shift intervals in Theorem 1.1 are consequences of the stated coercivity floors and the quadratic perturbation estimate; they are not asserted to be sharp, and they do not connect to the interior shift ranges studied in the companion manuscripts of this repository on total horizons at most $\log 2$ and $\log 3$.

For horizons beyond the radius of the present origin-centered profile, one can seek validated piecewise polynomial approximations to w on the positive real axis, retaining the logarithmic kernel exactly. The complex poles of G_0 lie at nonzero integer multiples of $i\pi$, whereas w is smooth at every nonnegative real argument. Taylor expansions about positive centers or Chebyshev pieces can therefore cover any fixed finite interval, provided their errors and derivative variations are enclosed. This removes the specific $L < 3$ profile restriction in principle; it does not prove coercivity at larger depth or bound the cost of doing so. The one idea that could change the scaling is a structured tail bound that keeps the prime–gamma cancellation instead of paying Σ_L in full, for instance on frequency bands; whether that is easier than the original problem is exactly the question, and nothing in the present data decides it.

A Moment formulas and assembly conventions

The code constructs all polynomial coefficients and integrates them against the moments below. For an integer $k \geq 0$, put $s = k + 1$, $H_s = \sum_{j=1}^s 1/j$, and $H_s^{(2)} = \sum_{j=1}^s 1/j^2$. The full-interval formulas are

$$\begin{aligned} \int_0^1 u^k du &= \frac{1}{s}, & \int_0^1 u^k \log u du &= -\frac{1}{s^2}, \\ \int_0^1 u^k (\log u)^2 du &= \frac{2}{s^3}, & \int_0^1 u^k \log(1-u) du &= -\frac{H_s}{s}, \\ \int_0^1 u^k \log u \log(1-u) du &= \frac{H_s}{s^2} - \frac{\pi^2/6 - H_s^{(2)}}{s}. \end{aligned} \tag{A.1}$$

They follow by differentiating the beta integral; the first three also follow directly by differentiating $1/s$. For $0 < \delta < 1$,

$$\begin{aligned} \int_{\delta}^1 u^k du &= \frac{1 - \delta^s}{s}, \\ \int_{\delta}^1 u^k \log u du &= -\frac{1 - \delta^s}{s^2} - \frac{\delta^s \log \delta}{s}, \\ \int_{\delta}^1 u^k \log(1 - u) du &= \frac{(1 - \delta^s) \log(1 - \delta) - H_s + \sum_{j=1}^s \delta^j / j}{s}. \end{aligned} \tag{A.2}$$

Integration by parts gives the last formula. Products of delayed polynomials use $\int_{\delta}^b u^k du = (b^s - \delta^s)/s$ on the appropriate nonempty window. For reflection, expand $(1 - u)^k$ by the exact binomial theorem.

To recover the gamma polynomial q_n in (6.3), multiply each coefficient of u^k in (6.1) by $\ell_L - H_k$, and add its contributions $g_j L^j (j - 1)! k! / (j + k)!$ at degree $k + j$. The beta factor is computed rationally with the recurrence $\beta_{1,k} = 1/(k + 1)$, $\beta_{j,k} = \beta_{j-1,k} (j - 1) / (j + k)$. The series g_j is obtained by reciprocal power-series inversion of $\sinh t/t$, multiplication by $e^{t/2}$, and subtraction of $4t \cosh(t/2)$, all over the rationals.

The unit-coordinate integrals acquire factors $\sqrt{(2i + 1)(2j + 1)}$ from normalization. The saved matrices use the causal derivative U . If instead $H_1 = UR_L$ is used, the two full-output Grams acquire the entry sign $(-1)^{i+j}$; this sign is 1 within either same-parity block. The central head and central leakage are unchanged.

B Algebraic identities for a possible depth recursion

The following identities are elementary and are recorded only because an induction in arithmetic depth would have to carry them. Nothing in Sections 6–7 depends on them.

An exact storage update with memory. Fix a positive shift and suppose $D_X = I - X^*X \succ 0$ and $\|Z\| < 1$, with bounded positive inverses. Put

$$W = I - Z^*Z, \quad \mathcal{C} = (I - ZZ^*)^{-1/2} Y D_X^{-1/2}, \quad S = D_X^{1/2} (I - \mathcal{C}^* \mathcal{C}) D_X^{1/2}.$$

For $V = \begin{pmatrix} X & 0 \\ Y & Z \end{pmatrix}$ one has the exact factorization

$$I - V^*V = \begin{pmatrix} I & -Y^*ZW^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & W \end{pmatrix} \begin{pmatrix} I & 0 \\ -W^{-1}Z^*Y & I \end{pmatrix}, \tag{B.1}$$

verified by multiplication using $I + Z(I - Z^*Z)^{-1}Z^* = (I - ZZ^*)^{-1}$. Thus contraction is equivalent to $\|\mathcal{C}\| \leq 1$; if $\|\mathcal{C}\| < 1$ the two triangular factors and S provide the complete inverse defect for the next append operation, while if only equality is certified the defect may be singular and an inverse-based recursion cannot automatically continue. A scalar lower bound on D_X discards orientation: for $Z = 0$ and $Y = (c, 0)$, the defects $\text{diag}(\varepsilon, 1)$ and $\text{diag}(1, \varepsilon)$ have the same least eigenvalue, but their updates are positive precisely when $c^2 \leq \varepsilon$ and $c^2 \leq 1$, respectively; $\varepsilon = 1/4$, $c^2 = 1/2$ gives a failing and a passing update with identical old scalar margins.

Nested Schur elimination for a structured tail. Split a form into an old head, a finite buffer, and the remaining tail,

$$Q_0 - mI = \begin{pmatrix} A & B^* & C^* \\ B & D & E^* \\ C & E & F \end{pmatrix}, \quad F \succeq \alpha I, \quad \alpha > 0.$$

Replacing F by αI and eliminating the tail gives $\hat{A} = A - C^*C/\alpha$, $\hat{B} = B - E^*C/\alpha$, $\hat{D} = D - E^*E/\alpha$, and the two tests

$$\hat{D} \succ 0, \quad \hat{A} - \hat{B}^* \hat{D}^{-1} \hat{B} \succeq 0 \quad (\text{B.2})$$

are sufficient for $Q_0 \succeq mI$, by two completions of the square. The cross-Gram C^*E must be retained; full-output parity Grams for the enlarged retained space supply C^*C , C^*E and E^*E . The blocks displayed here are those of the exact form Q_0 , whereas constructed Grams enclose the model form $\tilde{Q}$ of Lemma 4.4 in ball arithmetic; a certificate built on (B.2) would therefore have to carry the profile error η_M and the ball radii into the head, the buffer and every coupling product, as (5.4) does for the present two-block test. With the same enlarged space and scalar far-tail floor this is a reordering of the existing test, not a stronger theorem; it becomes one only with a proved structured lower bound for F .

C Reproducibility and review map

The repository folder `papers/weil-depth` contains the LaTeX source, bibliography, compiled manuscript, continuation guide, and the full numerical supplement under `numerics/`; older drafts and the reviews that produced each version are kept under `archive/`. The core numerical files are:

- `certify_arb.py`: exact-rational profiles, full-output ball construction, analytic tail bounds, and ball LDL;
- `enclose_certificate.py`: sector-wise floor bisection, ball Rayleigh upper bounds, and exact continuation constants, producing `enclosure.json` in each output folder and the summary table;
- `analyze_certificate.py`: hash-bound replay, rational continuation checks and the diagnostic relative coupling;
- `make_tables.py`: generates the tables of this paper from the saved records, so that no number in Tables 1–4 is transcribed by hand;
- `output/<horizon>_N128/`: for each of the seven horizons, the complete pivot lists with the archive hash and a content hash of the matrices (`central_certificate.json`), the enclosure record, and the continuation analysis. The saved ball matrices themselves (`central_matrices.json.gz`, about 13 MB per horizon) are deterministic output of the builder and are distributed separately rather than versioned; their expected layout and hashes are documented in `ARCHIVES.md`. In the tested environments (three Python versions with the same python-flint and FLINT releases) a rebuild reproduced the content hash exactly; the hash identifies the saved data and is not a claim of bit-for-bit portability across other arithmetic libraries or versions;
- `independent_arb.py`: the separately written construction from the printed formulas, with its pivot records at all seven horizons and saved matrices at 9/5 and log 7;
- `review_checks.py` and `review_claude/`: moment, arithmetic-generator, parity and failure controls, and the Fourier-side checks described in Section 6.

The certificates were produced with Python 3.10–3.12, python-flint 0.9.0 and FLINT 3.6.0; the two original horizons were rebuilt independently under Python 3.12.14 (original), 3.10.0 (version 0.2 review) and 3.11.15 (version 0.3 review) with identical pivots. NumPy is used only for diagnostic

eigenvalues and mpmath only for the candidate ground-state vectors and the review checks. Run from the numerical supplement directory, with assertions enabled:

```
python certify_arb.py --N 128 --M 220 --bits 1792 \
  --log-horizon 7 --floor 1.37e-28 --output output/log7_rebuilt
python enclose_certificate.py output/log7_rebuilt
python analyze_certificate.py output/log7_rebuilt \
  --shift 4e-15 --decay 6e-29 --generator-bound 14
python certify_arb.py --reuse --bits 1536 --floor 3.38e-23 \
  --output output/length_1p8_N128
python enclose_certificate.py output/log2_N128 output/log3_N128 \
  output/log4_N128 output/log5_N128 output/log6_N128 \
  output/length_1p8_N128 output/log7_N128 --summary output
python make_tables.py
```

A full rebuild checks the construction as well as the final sign tests; replay checks the signs from the saved construction enclosures. Do not run Python with `-O`. Build times are machine-dependent; each 128-mode construction took between 30 and 90 seconds, and each enclosure about one minute. The reviewed commands reject optimized Python mode and return a nonzero exit status if certification fails.

The largest mathematical review priorities are the normalization in (2.6) and (3.3), the weighted-line energy argument of Proposition 3.4, the operator-level fractional tail estimate, and the full-output parity identity. The numerical review should independently check the moment assembly, outward ball serialization, and the subtraction of (5.4) before each LDL test. No external certificate is an input to any proof here.

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